Angudubo, S[1]., Benavente, G[2]., van Etten, J[2]., & Mugisa, I[3]. (2026). Insights into the costing of on-farm testing: A case study of common bean tricot trials by the National Agricultural Research Organization (NARO).

[1]Alliance of Bioversity International and CIAT, Uganda
[2]Alliance of Bioversity International and CIAT, France
[3]National Agricultural Research Organization

1000FARMS

Insights into the costing of on-farm testing: A case study of common bean tricot trials by the National Agricultural Research Organization

Angudubo, S., Benavente, G., van Etten, J., & Mugisa, I. (2026).

Suggested citation:
Angudubo, S., Benavente, G., van Etten, J., & Mugisa, I. (2026). *Insights into the costing of on-farm testing: A case study of common bean tricot trials by the National Agricultural Research Organization (NARO).* Alliance of Bioversity International and CIAT.

Executive Summary

This report presents a comprehensive activity-based costing analysis of on-farm citizen science trials using the tricot (triadic comparisons of technologies) approach, focusing on a baseline case study of common bean (*Phaseolus vulgaris*) trials conducted by the National Agricultural Research Organization (NARO) at the National Crops Resources Research Institute (NaCRRI) in Uganda. Utilizing the 1000FARMS platform's standardized tricot costing framework, this evaluation dissects the financial and operational mechanics of decentralized, on-farm testing to establish a transparent baseline for institutional planning and future project scaling.

KEY OPERATIONAL PARAMETERS

The baseline evaluation models a structured seasonal trial consisting of 120 participating farmers evaluating 24 distinct bean varieties. Each farmer manages three decentralized plots under local agronomic conditions, yielding a total network size of 360 plots across the target trial landscape. The institutional setup involves a multi-tiered execution strategy combining specialized NaCRRI scientific researchers (Breeders, Research Officers, Research Assistants), technical support personnel, and decentralized extension field agents who handle direct, localized interactions.

CORE FINANCIAL STRUCTURE AND INSIGHTS

An analysis of the trial's financial architecture reveals a critical structural imbalance between centralized oversight and field execution across the primary cost pillars:

- **Labor Costs (37%):** Personnel wages are the largest driver of trial expenditures. This category is heavily dominated by the high internal daily rates of centralized institutional research staff rather than decentralized extension networks, highlighting a strong operational dependency on centralized supervision for routine trial management.
- **Travel expenses (27%):** Field logistics represent the true heavyweight of the operational budget, split between Per Diem (25%) and Travel Allowances (2%). Notably, centralized research team travel alone accounts for 51% of all travel expenses due to frequent, repetitive monitoring trips, severely limiting the program's geographic elasticity and scalability.
- **Material Costs (7%):** Physical trial inputs—encompassing seed preparation, packaging, communication bundles, and localized participant training supplies—account for a smaller share of the total expenditure. This demonstrates that field logistics and personnel allocation, rather than physical inputs, determine the trial's financial profile.

STRATEGIC COST DRIVERS AND SENSITIVITY

Sensitivity and scenario analyses demonstrate that the trial's financial sustainability is disproportionately governed by institutional behavior rather than material scale. Increasing the number of participating farmers introduces highly favorable economies of scale, dramatically reducing the per-farmer marginal cost. Conversely, minor increases in research team field monitoring frequencies or fuel price indices cause significant, compounded budgetary expansions due to the high variable cost of mobilizing centralized institutional staff.

KEY STRATEGIC RECOMMENDATIONS

To transform on-farm testing from an expensive, centralized project-based model into a lean, highly scalable national framework, this report recommends a paradigm shift across three core areas:

1. **Transition to Decentralized Local Delivery:** Institutionalize a localized operational model where routine data collection, localized seed distributions, and day-to-day farmer engagements are fully delegated to decentralized field agents and local organizations already operating within the target sub-counties.
2. **Implementation of Lean Research Supervision Protocols:** Restructure the centralized research team's role from operational execution to strict quality assurance. Centralized multi-role research caravans should be replaced by a single, agile trial manager utilizing a localized quality audit checklist (evaluating plot layouts, digital data entry integrity in ClimMob, and random verification protocols) during a heavily reduced calendar of field visits.
3. **Hub-and-Spoke Geographic Clustering:** Restructure distribution networks by clustering participating farmers into geographic hubs of 10-20 individuals each. Scheduling training, material payouts, and data collection hub-by-hub rather than farm-by-farm dramatically curtails logistical bottlenecks, minimizes vehicle wear, and leverages community cohesion to ensure long-term data quality.

By implementing these efficiency measures, NARO-NaCRRI can safeguard data integrity, reduce per-unit testing costs, and successfully scale the tricot approach across broader agro-ecological zones to deliver climate-resilient varieties to smallholders faster and more cost-effectively.

Introduction

COSTING ON-FARM TESTING

On-farm testing is essential for ensuring that agricultural innovations deliver value under the conditions where they are actually used. Performance observed on research stations often reflects optimal management, controlled environments, and uniform plots. In contrast, farmers operate under heterogeneous soils, microclimates, input constraints, labor trade-offs, and diverse market incentives. If varietal and technology evaluation does not involve these realities, recommendations risk being technically sound but contextually misaligned. On-farm testing, therefore, strengthens external validity: it captures genotype × environment × management (GxExM) interactions, reveals trait trade-offs that matter to farmers, and generates evidence that is directly relevant for adoption decisions. It also enhances legitimacy, as farmers become co-evaluators rather than passive recipients of innovation.

African public breeding programs have rarely performed on-farm testing of breeding products at scale, gathering sufficient data to make robust recommendations for variety release and promotion. To address this situation, the 1000FARMS platform was launched in 2021 with funding from the Bill & Melinda Gates Foundation and is implemented through partnerships led by the Alliance of Bioversity International and CIAT, together with IITA, CIMMYT, Wageningen University, and INN University (until 2023). The project works closely with CGIAR-NARES breeding networks and links to other CGIAR-led initiatives to support national testing and scaling efforts. The 1000FARMS platform aims to establish an Africa-wide network for on-farm testing for better breeding decisions and variety placement recommendations.

One limitation of conventional on-farm testing methods is their scalability. The tricot (triadic comparison of technologies) approach provides a scalable and statistically rigorous way to implement decentralized on-farm testing. Instead of testing many varieties in a few locations, tricot distributes small, incomplete blocks—typically three options—to many farmers. Each farmer ranks performance under their own management, while advanced statistical models aggregate these comparisons to estimate overall and segment-specific performance. This design reduces logistical complexity, increases geographic coverage, and allows credible inference across the target population of environments. By combining farmer-generated data with robust analytics, tricot bridges scientific rigor and real-world relevance, strengthening variety release, product design, product placement, and seed system decisions.

Scalability also depends on the cost per data point of on-farm testing. The costs of on-farm testing depend heavily on the operational decisions made by the implementers. To better understand the drivers of on-farm testing costs and refine our costing methodology, we conducted a series of case studies. This study focuses on the common bean in Uganda.

CASE STUDY CONTEXT: NARO

The National Agricultural Research Organization (NARO) is Uganda's apex institution for publicly funded agricultural research. Established under the National Agricultural Research Act (2005), NARO coordinates, oversees, and conducts research across crops, livestock, fisheries, forestry, and natural resources. NARO is one of the national institutions collaborating with the Alliance of Bioversity International and CIAT through the CGIAR-NARES network, including the 1000FARMS platform.

This case study focuses on NARO's common bean tricot trial implemented through the National Crops Resources Research Institute (NaCRRI), one of NARO's Public Agricultural Research Institutes (PARIs). The trial aimed to generate farmer-led evidence on varietal performance and preferences under real farm conditions, to inform variety advancement and dissemination decisions in the legumes research program.

With support from the Accelerated Variety Improvement and Seed Systems in Africa (AVISA) project and the 1000FARMS platform, NaCRRI established the trials in the first production season of 2025 (March – July). A total of 120 tricot trials were established in two agro-ecological zones: the Lake Victoria Crescent (Lwengo and Nakaseke districts) and the Mid-Western zone (Hoima district). The trials assessed four common bean product profiles: yellow, sugars, red mottled/red, and white.

NaCRRI legumes program selected the tricot approach because it enables decentralized testing at scale while maintaining structured comparison of technology options across diverse environments. Under tricot, each participating farmer manages and evaluates three randomly assigned varieties drawn from a larger set of candidate lines. Farmers provide comparative rankings that complement agronomic assessments and help identify varieties that fit target environments and user preferences. By strengthening the link between varietal evaluation and farmer priorities, the approach can improve the relevance and uptake of common bean varieties.

This report documents the methods, parameters, and assumptions used to estimate the implementation costs for the NaCRRI common bean tricot trials. It presents the major expenditure components, quantifies each component's contribution to total costs, and identifies the main cost drivers. The report examines how deviations from standard tricot steps affect cost-effectiveness and operational efficiency. The analysis links the costs to operational decisions, including staff role allocation, travel intensity, and delivery model. Further, the report discusses what these results imply for scaling decentralized on-farm testing within NARO's legumes program. It concludes with recommendations to reduce avoidable costs, ensure data quality, and support the institutionalization of tricot-based on-farm testing within Uganda's National Agricultural Research and Extension Systems (NARES).

COSTING OBJECTIVES

Systematic costing estimates the resources needed to implement a trial as unit costs per data point (per farmer or per plot/block). However, the benefit of costing goes beyond obtaining the numbers. Costing helps programs understand how operational decisions, such as who does what, where, and how often, translate into resource requirements, such as time, money, and logistical demands. These decisions have consequences for data completeness and quality, farmer engagement, and the feasibility of running the trials. By making costs and their trade-offs explicit, programs can assess operational decisions or test scenarios in a transparent way and compare them across partners and locations. Such comparisons support learning, improve planning, and strengthen decisions about scaling. In summary, costing helps with the following objectives.

1. Supporting realistic budgeting and resource mobilization;
2. Identifying efficiency gains without compromising data quality or partnership quality;
3. Enabling fair and sustainable cost-sharing among research, extension, and seed system stakeholders; and
4. Allowing for learning and improving delivery models across institutions and regions.

Costing Methodology and Assumptions

The tricot costing is prepared using the activity-based Excel Costing Template (https://hdl.handle.net/10568/182169) following the procedures outlined in the "Tricot Costing Template Guidelines" (https://hdl.handle.net/10568/181933). In particular, the guidelines specify that:

- Travel expenses are estimated in accordance with United Nations travel reimbursement standards (https://hr.un.org/node/53991).
- Communication costs follow country-level ICT price baskets as published by the International Telecommunication Union (ITU) (https://www.itu.int/en/ITU-D/Statistics/Dashboards/Pages/IPB.aspx).
- Unit prices for materials and services are based on market prices in each country and are adjusted for inflation where needed.

The key assumption is that the legumes breeding program implemented tricot trials as prescribed, following the ten steps outlined in "The Tricot Guide" (https://hdl.handle.net/10568/109942).

Other assumptions include:

- The research team leads the initial work: trial preparation, experimental design, and seed packaging.
- Farmer-facing organizations handle most implementation activities because they work directly with farmers. The research team focuses on quality assurance through spot checks to support trial implementation under real farm conditions.
- Field agents from farmer-facing organizations receive sufficient training and then independently oversee field implementation.
- NaCRRI plans seed production for tricot trials separately from other breeding trials to ensure the required quantities are available on time.
- The research team distributes seed centrally to agreed drop-off points, and field agents from farmer-facing organizations complete the last-mile delivery to participating farmers.
- Field agents lead all monitoring and data collection. The research team supports specialized tasks, including yield measurement (covering ~30% of farms) and final farmer feedback workshops.
- The central breeding budget covers the research team's time for trial preparation, seed packaging, and field agent training. While 10–15% of researchers' staff costs are allocated to on-farm testing.
- The costing uses institution-specific daily and hourly labor rates, calculated from official monthly salary scales for each job role within NARO.

Cost Structure and Components

This section summarizes the trial's cost components and provides insights into stage- and role-level cost patterns. Table 1 presents the structure of the marginal trial costs.

Table 1: Tricot implementation by component

Cost component	Sub-category	Trial cost (USD)	Share of total cost (%)	Cost/ farmer (USD)
Travel expenses	Per diem	2,764.8	25.0	23.04
	Travel allowance	255.4	2.3	2.13
Material	-	702.8	6.3	5.86
Labor	Field-based	4,069.0	36.8	33.91
Transport	Fuel & O/M	3,236.7	29.2	26.97
Communication	-	41.4	0.4	0.35
Total		**11,070.0**		**92.25**

KEY OBSERVATIONS

Table 1 summarizes the marginal (variable) implementation costs for the NaCRRI common bean tricot trial. These are costs that increase as the legumes program recruits more farmers because they relate to field delivery. In this costing, office-based activities are considered as fixed costs. The research team incurs these costs during trial preparation and coordination of on-station activities, and they do not change as the number of farmers increases. The costs cover meetings for selecting test lines and traits for evaluation, planning trial establishment, trial design, testing, and deployment on ClimMob, and centralized training for field agents. All research team staff time spent on trial preparation tasks is treated as a fixed cost. When field agent training is organized once at the institute (the trial coordination center) for the entire farmer network, it is also treated as a fixed cost because it is not repeated for each farmer. This choice increases the total fixed costs but lowers the marginal cost per additional farmer. The fixed costs are excluded from the results in Table 1.

The costing summary of the NARO/NaCRRI common bean tricot trial shows that travel expenses, field-based labor, and transport are the largest cost components. Together, these components total USD 10,070.4, representing about 91% of total marginal costs. The breakdown of these components is presented below:

- **Labor costs:** Field-based labor is the largest cost component, accounting for about 37% of total marginal costs. It covers staff time for training field agents, distributing seed, monitoring, data collection, yield measurement, product evaluation, and farmer feedback. Monitoring constitutes the largest sub-category under this component, accounting for 86% of field-based labor costs. Most of the monitoring costs are concentrated in the two research team visits, which account for 77% of the monitoring labor costs while reaching only about 25% of participating farmers. Monitoring costs are high because multiple research roles (breeder, research officer, and research assistant) travel together for routine follow-up visits that trained local field agents could conduct at lower cost, with the research team shifting its focus to quality assurance.

- **Transport costs:** In the NARO/NaCRRI common bean tricot trial, transport expenses, covering fuel and vehicle operation/maintenance, represent the second-largest cost component (29%). These costs mainly arise from field activities that require the research team to travel, including seed distribution, follow-up visits, and yield measurement.

  Transport costs increase during seed distribution because the current delivery model relies on centralized vehicles and drivers to deliver seed packages to field agents' locations. In some cases, the program combines delivery with planting support to ensure quality, further increasing kilometers traveled and time on the road. Transport costs also remain high during monitoring and yield measurement, even though the research team visits only about 25% of participating farmers. This suggests that the sampling and routing of visits do not exploit geographic clustering. Random selection of farmers without clustered route planning results in long travel distances between dispersed sites, increasing costs per visit.

- **Travel expenses are the third-**largest cost component, accounting for 27% of total costs. These costs mainly reflect per diem and travel allowances. Research team travel accounts for 51% of all travel expenses. Most per diems are related to the research team's involvement in routine field activities, such as monitoring, yield measurement, and product evaluation, which require repeated visits to trial sites. This pattern indicates that travel expenses increase when staff take on tasks that could be delivered through trained local field agents with limited audit-based supervision by the research team (e.g., spot checks and remote quality verification).

- **Material costs accounted for only 6% of total costs in the NaCRRI common bean tricot trials, indicat**ing that inputs and supplies were not the main cost drivers. This reflects the design logic of tricot, in which trials are implemented under farmer-managed conditions rather than fully research-managed ones. The trial seed accounted for only 16% of total material costs, suggesting that seed multiplication is not as costly as often assumed relative to the full implementation cost. With appropriate planning, the number of participants and seed requirements can be estimated early using a ClimMob-based calculator, helping teams produce adequate seed quantities in a cost-efficient way. Fertilizers and pesticides were not supplied because their use varies across farmers. Some farmers apply them, while others do not, depending on access, cost, or routine farm management practices. This decision helped maintain realistic farmer conditions and avoided introducing input-use differences that would not reflect normal production practices.

  However, sisal rolls used for field marking and row planting accounted for about 35% of total material costs. Unlike fertilizers or pesticides, sisal strings were intentionally provided to field agents to support proper plot layout. The aim was to encourage farmers to plant in lines rather than broadcast the seeds, while keeping the trial simple and farmer-managed. This cost was linked to trial standardization and data quality, not the usual input use. In future trials, this cost could be reduced by using locally available row markers where appropriate, thus maintaining the benefits of row planting while minimizing material expenses.

  A second material-related cost is participant support, mainly transport refunds for activities that require farmers to meet at a central venue (e.g., training, seed distribution, or product evaluation). These costs rise when meetings are organized frequently or when participants are not clustered into groups. In practice, NaCRRI can keep material costs low by (i) aligning trial establishment with real farm practice and (ii) reducing the number of centralized gatherings through clustering farmers into groups (e.g., 10-20 farmers per hub).

Cost Drivers in the Common Bean Tricot Implementation

In tricot trials, costs depend largely on operational choices and decisions, including how the implementing team designs the delivery model, schedules field activities, assigns roles, and scales farmer participation. In the NARO/ NaCRRI common bean tricot trials, the largest cost components- staff time (labor), transport, and travel-related expenses- reflect these implementation decisions. They increase when activities are delivered through repeated in-person visits, when researchers join routine field activities that field agents can handle, and when farmers are served across wide geographic areas without geographic clustering. Costs also rise when teams conduct activities such as separate trips instead of bundling them into fewer, well-planned field routes with fixed schedules.

The first cost driver in the common bean tricot trial is the operational decision to deploy multiple research roles for routine field delivery. Researchers participate directly in activities that local field agents can deliver at scale, including field agent training, monitoring, yield measurement, product evaluation, and farmer feedback sessions. In this trial, the researchers reached only about 25% of farmers during routine activities, while training and farmer feedback sessions may involve all the farmers. This creates overlapping field presence, where researchers and local field agents often travel to the same sites to conduct related tasks. As a result, each additional researcher trip involves staff time, transport, and per diem costs. This duplication appears to increase labor, travel, and transport cost components, without a proportional improvement in the evidence generated from the tricot trial.

The first cost driver greatly influences field-based labor costs because costs depend not only on how many activities are conducted, but also on who delivers routine activities and how often they travel. When multiple research roles participate in routine fieldwork, the number of paid staff days increases, even when tasks per farmer are simple and repeatable. A practical gain comes from delegating routine tasks, such as monitoring, data collection, and yield measurement, to trained local field agents, covering 25% of farmers whom researchers usually cover. This operational shift reduces research staff days and associated travel time, while shifting their focus to planned spot checks and data audits. Costing the bean tricot trial under this assumption would reduce field-based labor costs by about 81%, while maintaining technical oversight of evidence quality and avoiding staff time costs associated with travel.

The first cost driver also significantly influences transport costs. Each additional field trip by the research team to conduct routine field activities increases fuel costs, vehicle operation, and maintenance costs. This pattern increases the transport cost component even when travel distances remain the same. Shifting routine field activities to trained local field agents while the research team focuses on planned spot checks and data audits would reduce transport costs by about 31%. However, decentralization should be supported by careful selection of field agents, practical training, simple data quality checks, routine spot checks, clear accountability mechanisms, and timely feedback loops. This will help ensure that cost savings do not come at the expense of data quality.

A second major cost driver in the common bean tricot trial is the delivery model, which decentralizes both field agent training and seed distribution and engages farmers through many small meetings. Training field agents in the field, rather than through a single centralized session at the coordination center, increase the number of trips and raise both transport costs and travel-related expenses. Similarly, seed distribution to lower administrative units (e.g., at the parish level) results in multiple delivery routes, increasing driver time and transport costs. Furthermore, organizing many small-farmer meetings during product evaluation and farmer feedback increases travel frequency and associated

costs. Overall, the cost increase comes from repeated travel and logistics for activities that could be bundled into fewer, well-planned sessions.

Shifting key operations into a more centralized delivery model can reduce travel and transport costs. Field agents can be trained at the coordination center, and seed packages can be delivered in bulk to a single hub location. Local field agents can then complete the last-mile distribution of packages to farmers in their locations, reducing the need for repeated trips by the coordination team. Second, product evaluation and farmer feedback sessions can be organized in clusters. Holding fewer, larger sessions reduces per diem and transport costs, while maintaining participation and data quality.

Scenario analysis shows that small changes in field delivery can reduce transport and travel-related costs without affecting tricot procedures. First, organizing field agent training at the coordination center reduces travel expenses by about 10% and transport costs by about 13%, mainly by eliminating repeated trips for the same purpose. Second, reducing kilometers traveled during seed distribution by about 60%, by delivering seed packages to central drop-off points and allowing local field agents to handle last-mile delivery, reduces transport costs by about 8%. Third, reducing meeting sessions by 50% through clustering farmers into groups (for example, 10–20 farmers per hub) during product evaluation and farmer feedback lowers transport costs by about 11%.

The largest savings come from combining shifts in operational decisions. Centralization of activities, clustering farmers, and bundling tasks, such as merging farmer training with seed distribution in the same visit, reduces duplication of trips, improves scheduling, and increases the cost savings beyond what each measure achieves on its own. Overall, a fully centralized, cluster-based delivery model for training field agents, seed distribution, product evaluation, and farmer feedback would reduce travel expenses by reducing separate trips and focusing on fewer locations. Under this assumption, travel expenses are reduced by about 10%, while transport costs decline by about 32%. These savings come mainly from reduced travel distances and more efficient routing, rather than changes in staff time.

Finally, it is important to distinguish between the cost of delivering on-farm testing and the cost of routine breeding work. A key budgeting decision concerns what share of researchers' staff time should be charged to on-farm testing. For the common bean tricot trial, it is reasonable to treat office-based activities, such as trial preparation, field agent training, and seed package preparation, as part of the core breeding budget. These tasks are standard functions of a breeding coordination center, and when the NaCRRI team delivers them at the center, it avoids travel-related marginal costs.

Under this framing, on-farm testing should carry only the researcher time needed for essential, project-supported supervision and quality assurance. A small supervision share is therefore justified (e.g., 10% as proposed by the ABI Transform team or 15% as applied in the NARO common bean tricot costing), while the remaining researcher time remains under central breeding budget. This approach does not change the underlying field delivery costs; it improves budget transparency by avoiding double-counting, makes cost comparisons across trials and institutions more consistent, and clarifies which costs are financed by projects and which are covered by routine breeding resources.

A scenario analysis of NARO's common bean tricot trial shows that a small set of operational choices largely determine costs. Costs increase rapidly when routine field activities rely on repeated researcher travel and many small, duplicated meetings, because each additional farmer adds to coordination time, per diem costs, and transport costs. Costs decline when routine activities, such as monitoring and data collection, are delivered through local field agents, while research shifts to targeted supervision through spot checks, remote support, and planned quality audits. Under this delivery model, the trial maintains data integrity, reduces travel-related expenses, and scales more easily.

Delegating routine monitoring, data collection, and yield measurement to trained local field agents reduces field-based labor costs. In this scenario, the cost per farmer falls from USD 92.25 to USD 63.97, and total costs drop by about 31%, saving USD 3,393.7. The same operational shift also reduces travel needs by limiting repeated researcher travel. Travel savings are small but still meaningful: the cost per farmer decreased from USD 92.25 to USD 83.90, and total costs fell by 9%, saving USD 1001.8.

When the legumes program applies both changes together, local field agents deliver routine activities and the research team shifts focus to quality assurance, the combined effect is larger. The cost per farmer declines from USD 92.25 to USD 58.81, and total costs drop by 36%, saving USD 4,012.8. These results suggest that the program can increase the number of farmers without a proportional increase in costs, because routine field delivery no longer depends on accumulating staff days or on repeated travel by multiple research roles.

Shifting from decentralized delivery to a more centralized approach by combining training and seed distribution on the same schedule and clustering farmers for joint sessions reduces transport costs and research travel expenses. Under this scenario, the cost per farmer drops from USD 92.25 to USD 81.32, and total costs decrease by 12%, generating an estimated saving of USD 1,311.2. The savings come mainly from fewer trips and fewer separate meetings with farmers, while keeping the same training and seed distribution outputs.

Taken together, these scenarios show that the main constraint to scaling tricot is not the approach itself, but the delivery design. If the NaCRRI legumes program adopts local delivery as the default, centralizes training and seed distribution into fewer, well-planned sessions, and uses clustered engagement of farmers to reduce meeting and travel costs, the program can lower the cost per farmer from USD 92.25 to USD 47.40, and decrease total costs by 49%, resulting in saving of about USD 5,382.2, and expand coverage while maintaining data quality and successful completion of the trial cycle.

Conclusions and Recommendations

CONCLUSIONS

The NARO/ NaCRRI common bean tricot trial costing shows that decentralized on-farm testing using the tricot approach is cost-effective and can generate useful evidence for breeding and variety placement decisions. The main constraint is not the amount of data to be collected, but rather current operational decisions that deviate from core tricot principles. Costs increase mainly because implementation relies on repeated field travel, especially by the research team, limited delegation of routine field activities, and many small, fragmented meetings. If delivery is streamlined and responsibilities are clearly assigned in line with a crowdsourced, citizen science model, NARO/ NaCRRI can achieve the same trial outputs at a cost that remains sustainable as the program scales to more farmers.

RECOMMENDATIONS

Leverage local field agents for delivery of field work: The NaCRRI legumes program should delegate the delivery of routine activities, such as monitoring, data collection, and yield measurements, to trained local field agents as the standard operating model. The research team should limit co-travel for routine field visits. Instead, the researchers should focus on quality assurance through spot checks, remote verification, and data quality audits. This operational shift reduces the associated field-based labor, transport, and travel expenses by reducing the research team's field days and travel while maintaining technical oversight and data integrity.

Replace multi-role research field visits with a lean supervision protocol: Critical field supervision should shift from traveling in teams composed of multiple job roles (breeder, research officer, research assistant) to a lean structure led by a designated agile trial manager. The trial manager would conduct periodic spot checks using a standard quality assurance checklist to verify plot layout, variety allocation, seed package integrity, plot labeling, the farmer's understanding of trial instructions, planting and management records, and the completeness and consistency of data submitted through ClimMob. These audits should be organized around clustered routes and target a sample of farmers (e.g., 20–30%) to verify trial establishment, compliance with protocols, and key measurements. This approach reduces field-based labor, transport, and per diem costs while maintaining trial integrity, data quality, and the core principles of the tricot approach.

Centralize training of field agents: Training of field agents supporting the trial network should be conducted once at the coordination (NaCRRI) using standardized training modules. These can then train the farmers on the tricot approach, plot establishment, field management, and data collection. Field-based training should be limited to exceptional cases and bundled with other activities. Centralized training reduces repeated travel by the research team, lowers transport and per diem costs, and improves standard of implementation across locations.

Deliver seed packages in bulk to hub drop-off points: The legumes team should deliver seed packages to a limited number of predefined hub locations (e.g., sub-county or district hubs). The local field agents should be assigned to handle last-mile delivery to participating farmers in their area of operation. This operational shift reduces kilometers traveled and driver time during seed distribution and lowers fuel and vehicle operation costs, while maintaining timely seed delivery.

Cluster farmer engagement to reduce staff time and travel expenses per farmer during meetings: All farmer-facing activities that require gatherings, such as training, product evaluation, and farmer feedback sessions, should be organized through clusters of about 10–20 farmers per hub. All farmer engagement activities should be scheduled hub-by-hub rather than farm-by-farm across dispersed locations. Clustering reduces travel distances per activity, improves scheduling, and lowers both transport and travel-related expenses.

While farmers may already be meeting in groups of 15–20, the key improvement is to treat each group as a functional tricot hub, not just a meeting size. Each hub should bring together geographically close farmers, can be reached in one field circuit, and can be supported by the same field agents. Farmer engagement activities should therefore be scheduled hub-by-hub rather than farm-by-farm across dispersed locations. This reduces travel distances per activity, improves coordination, and lowers both staff time and travel-related costs. It also improves farmer learning by allowing nearby farmers to compare experiences and support each other during the trial.

Bundle farmer-facing activities into fewer visits and meeting sessions: Training of farmers, seed distribution, and trial establishment support should be scheduled as a single activity per hub, where possible. Similarly, the program can organize product evaluation and farmer feedback as fewer, large hub sessions rather than many small meetings. This change reduces repeated trips, lowers per diems, and maintains participation in and outputs of the meetings.

Institutionalize delivery model through SOPs and role guides: NaCRRI legumes program should formalize the above changes into a short set of standard operating procedures: role allocation (who does what), hub clustering rules, audit sampling and checklists, seed delivery steps, and meeting bundling guidance. Clear SOPs accompanied by implementation agreements or memoranda of understanding (MOUs) reduce ad hoc decisions that recreate travel, labor, and meeting costs as the tricot trial scales.

If NaCRRI adopts local delivery as a default operational model for routine activities, limits research involvement to planned quality assurance, and organizes farmers in geographic clusters, the program can greatly reduce field-based labor, transport, and travel expenses while maintaining data quality. This shift maintains the tricot procedures, whereby many farmers generate evidence through simple, repeatable steps under real farm conditions, while researchers safeguard quality through targeted audits rather than repeated travel. As duplication declines and activities are bundled around clustered hubs, the cost per farmer falls, and the trial becomes easier to scale.

Acknowledgements

We thank Rachel Chase for the support in developing the tricot costing template and for the feedback that strengthened this report. We also thank Dr. Immaculate Mugisa for coordinating field implementation, supporting costing, and providing data inputs from the NARO/NaCRRI legumes program. We are grateful to the NaCRRI legumes program technicians and field agents from farmer-facing organizations for their time, records, and practical insights, which enabled us to prepare this report.